

Unit 7, Lesson 12: Dividing Integers Using Integer Chips

Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can use integer chips and patterns to discover rules for dividing integers.

7.12.1 Warm-Up: Notice and Wonder

1. Write down one thing you notice about the picture.

2. Write down one thing you wonder about the picture.

3. Isaac described the image as a math fail. Why do you think he described it that way?

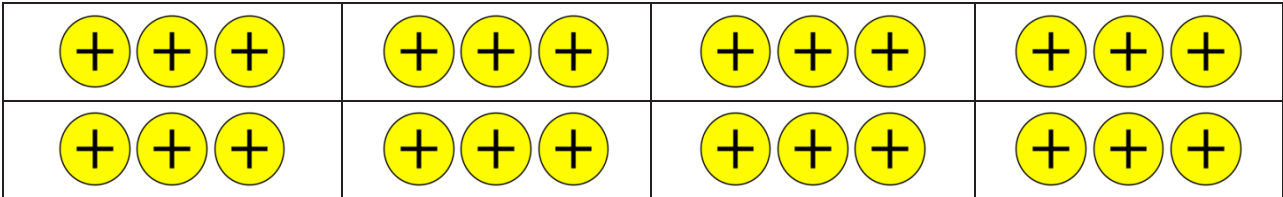
7.12.2 Guided Instruction: Using Integer Chips to Represent Division

Recall that division is repeated subtraction.

For example, $24 \div 8$ can be expressed as $24 - 8 - 8 - 8$ because 3 groups of 8 can be subtracted from 24.

This shows that 24 is made up of ____ groups of 8.

The expression $24 \div 8$ can also be interpreted as dividing 24 into 8 equal-sized groups. In the representation shown, 24 positive integer chips are divided into 8 equal-sized groups.



The quotient is the result of the division, or the value of the integer chips in each group.

$24 \div 8 = \underline{\hspace{2cm}}$

1. Complete the table to describe the division, then draw the visual model using integer chips, and determine the quotient.

Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{15}{5}$	Divide ____ into ____ equal-sized groups		

$-6 \div 2$	Divide ____ into ____ equal-sized groups		
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When modeling division with integer chips, the divisor must be positive, as this indicates the number of groups needed. When the divisor is negative, the expression can be rewritten using opposites.

For example, $\frac{8}{-2}$ can be rewritten as $\frac{-8}{2}$ because $\frac{8}{-2} = \frac{-8}{2}$.

2.

Division Expression	Equivalent Division Expression	Written Description	Visual Model of Division	Quotient
$\frac{9}{-3}$		Divide ____ into ____ equal-sized groups		
$-12 \div -2$		Divide ____ into ____ equal-sized groups		

3. Use the pattern of the division in the tables to complete the rules by drawing the sign of the quotient as an integer chip.

Visual Model of Division	Sign of Quotient
+ +	
+ -	
- +	
- -	

7.12.3 Collaboration: How Are Integer Multiplication and Division Related?

Jeannie and Solimar were discussing the relationship between $5 \times -2 = -10$ and $(-10) \div (-2) = 5$.

Solimar said, "The multiplication equation can be interpreted as 5 groups of -2 , and the division equation can be interpreted as how many groups of -2 are in -10 . Isn't that basically the same thing?"

Jeannie replied, "I agree. I also think it is interesting to see how the rules seem to be related. In the multiplication equation, we are multiplying a positive by a negative and the answer is a negative. In the division equation, we do the reverse of the multiplication problem and find that a negative divided by a negative is a positive."

Solimar said, "Yeah, that's a cool connection. Do you think the rules are the same for multiplication and division?"

Jeannie said, "I think they might be related to each other."

1. Discuss Solimar and Jeannie's conversation with your partner.
2. Create two integer division equations that can be rewritten with the related multiplication equation.

Division Equation	Related Multiplication Equation

7.12.4 Bring It Together: How Are Integer Multiplication and Division Related?

1. Write the missing factor in each given equation. Then, write the related division or multiplication equation.

Given Equation	Can be Thought of as...	Related Equation
$3 \times \underline{\hspace{1cm}} = 21$		$21 \div 3 = \underline{\hspace{1cm}}$
$\frac{-48}{4} = \underline{\hspace{1cm}}$		$(\underline{\hspace{1cm}})(4) = -48$
$-8 \times \underline{\hspace{1cm}} = 56$		$56 \div (-8) = \underline{\hspace{1cm}}$
$20 \div \underline{\hspace{1cm}} = -5$		$(-5)(\underline{\hspace{1cm}}) = 20$
$-\left(\frac{36}{4}\right) = \underline{\hspace{1cm}}$		$\underline{\hspace{1cm}} \times 4 = -36$

2. How are rules for multiplication of integers related to the rules for division of integers?

7.12.5 Collaboration: Which One Doesn't Belong?

1. Circle the expression that does not belong with the other three.

$$\frac{10}{-5}$$

$$\frac{-10}{5}$$

$$\frac{-10}{-5}$$

$$-\frac{10}{5}$$

2. Explain why the expression does not belong.
3. Verify your answers with your partner and, if necessary, resolve any differences.

Quotients can have negative signs in different places.

4. The values $\frac{12}{-2}$, $\frac{-12}{2}$, and $-\left(\frac{12}{2}\right)$ are equivalent. Explain why they are equivalent using your understanding of dividing integers.
5. Discuss your explanation with your partner. If necessary, edit your explanation based on your discussion.
6. Create two equivalent expressions for $21 \div (-7)$.
7. $\frac{-40}{-8}$ is not equivalent to $-\left(\frac{40}{8}\right)$. Explain why not.

7.12.6 Guided Practice: Dividing Integers

1. Find each quotient.

a. $12 \div -4 = \underline{\hspace{2cm}}$

b. $\frac{-63}{-9} = \underline{\hspace{2cm}}$

c. $-\left(\frac{28}{4}\right) = \underline{\hspace{2cm}}$

d. $60 \div 5 = \underline{\hspace{2cm}}$

e. $-\left(\frac{-30}{-5}\right) = \underline{\hspace{2cm}}$

2. Create two expressions equivalent to $\frac{-20}{10}$.

3. What is the value of the expressions above?

7.12.7 Your Turn!

1. In each column below, create three different division expressions that would result in the given quotient.

-4	10	-7
1.	1.	1.
2.	2.	2.
3.	3.	3.

7.12.8 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with understanding and using the rules for multiplying and dividing integers.



